EXPERIMENT 4

Academic Year: 2025-26 **Year and Sem: BEVIII**
Subject: Recommendation Systems Lab **Course Code: CSDOL8022**

Aim: To implement content-based recommendation system using Python
Theory: A Content-Based Recommendation System is a type of recommendation algorithm that recommends items to users based on the features or attributes of the items themselves, rather than on the preferences or behaviors of other users. Essentially, it suggests items that are similar to ones the user has shown interest in, using the characteristics of the items to make predictions.

How Content-Based Recommendation Systems Work

The basic idea is to recommend items that are similar to items a user has already interacted with, based on the content (features) of the items. These systems analyze the attributes of items (such as product descriptions, movie genres, keywords, etc.) and compare them to items the user has rated highly or interacted with in the past. Here's how it typically works:

1. Item Representation:
   Each item (e.g., movies, products, articles) is described by a set of features.
   For example:
   ■ Movies: genre, director, actors, plot summary
   ■ Products: brand, category, price, specifications
   ■ Articles: keywords, topics, author
   These features are often converted into numerical representations, such as vectors, to make them comparable. TF-IDF (Term Frequency-Inverse Document Frequency) or Word2Vec (for textual data) is commonly used for feature extraction.
2. User Profile:
   The system keeps track of a user's interactions with items (such as what items they've viewed, liked, rated highly, or purchased).
   A user profile is built based on these interactions, where the profile represents the preferences of the user in terms of the content features of the items they've interacted with.
3. **Similarity Calculation**

After converting items into vectors, the system calculates similarity between:
User Profile ↔ Other Items
Similarity measures used:

- Cosine Similarity
- Euclidean Distance
- Jaccard Similarity

Cosine Similarity Formula

$$Similarity = \frac{A \cdot B}{|A||B|}$$

Where
A = user profile vector
B = item vector

Range:

- 1 → highly similar
- 0 → not similar

4. **Recommendation Generation**
   Based on the calculated similarity scores, the system recommends items that have high similarity to the ones the user has interacted with. The user receives suggestions for new items that are most like the items they've already enjoyed.

Applications

Content-based recommendation systems are widely used in:

- Movie streaming platforms
- E-commerce websites
- News portals
- Music streaming services
- Online shopping systems

Conclusion

A content-based recommendation system leverages the features of items themselves to make personalized recommendations for users. It compares items based on their attributes (like genre, description, or price) and suggests similar items to those the user has already shown interest in.